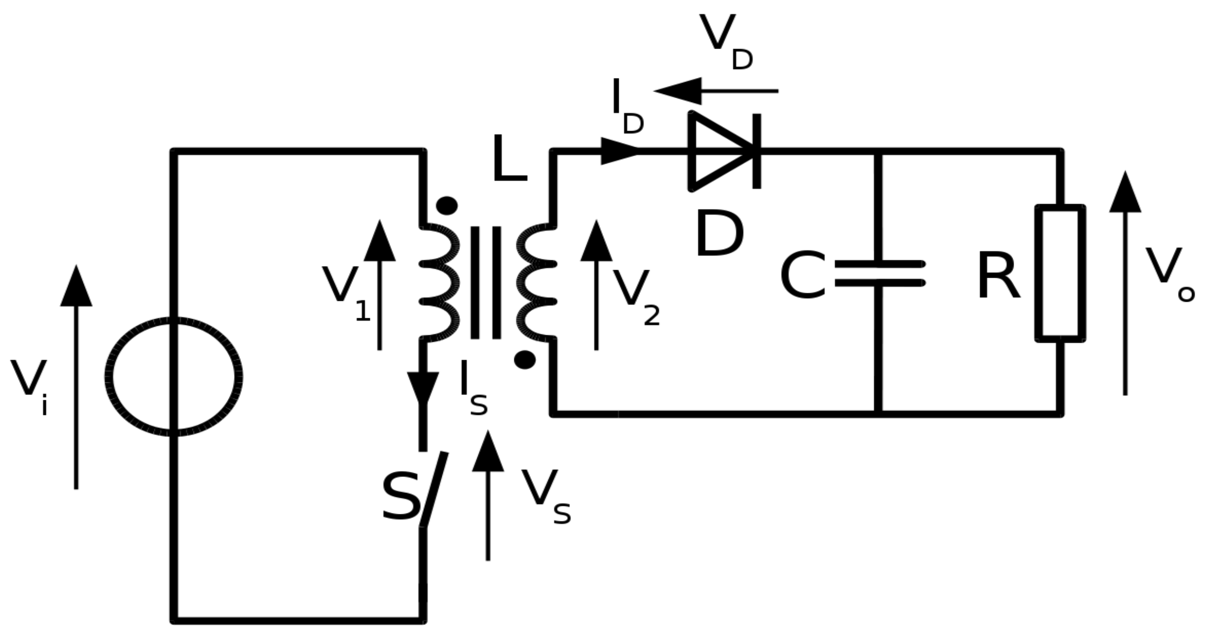
Flyback converter

Dc-Dc converter

What is a Flyback Converter?

The flyback converter is a power supply topology that uses mutually coupled inductor, to store energy when current passes through and releasing the energy when the power is removed. The flyback converters are similar to the booster converters in architecture and performance. However, the primary winding of the transformer replaces inductor while the secondary provides the output. In the flyback configuration, the primary and secondary windings are utilized as two separate inductors.



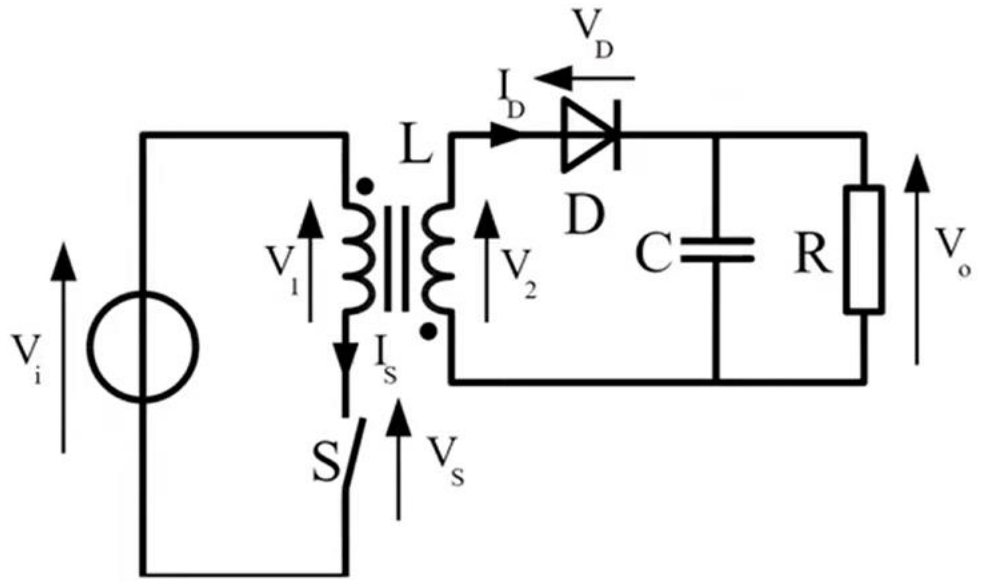
Applications of flyback converters

The flyback converter are used for a wide range of electronics applications such as:

-Television sets which consume small amount of power of up to about 250W
-Standby power supplies for computers
-Cell phone and mobile device chargers
-High-voltage supplies in TV and Monitor CRTs, Lasers, Xenon flashlights, copiers, etc.

Advantages of flyback converter

-The primary is isolated from the output.
-Capable of supplying multiple output voltages, all isolated from the primary.
-Ability to regulate the multiple output voltages with a single control.
-Can operate on a wide range of input voltages
-The Flyback converters use very few components compared to the other types of SMPSs.



Parameters Provided

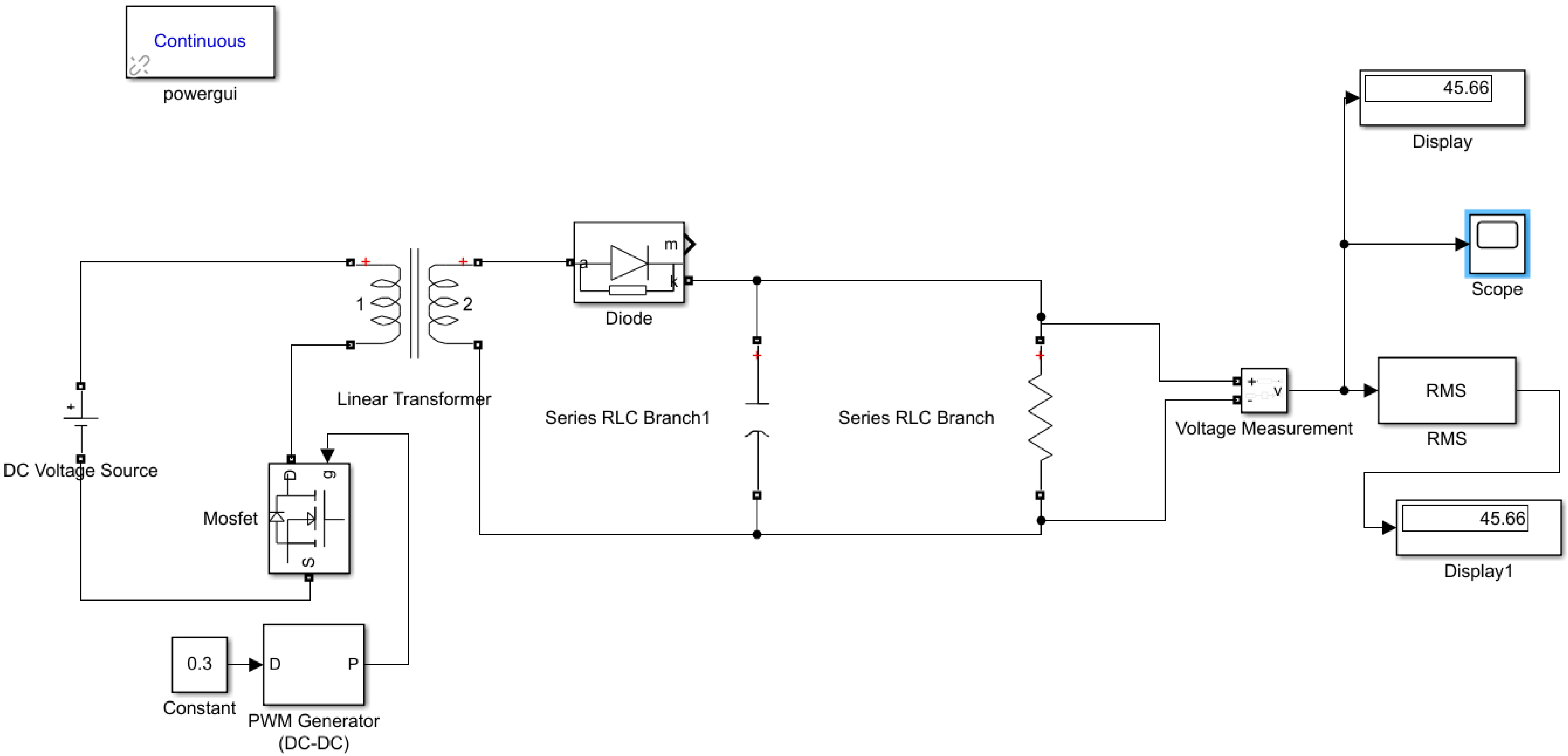
Vo = 48 V, Vs = 24V, P = 50W, R = 50ohm, Switching Frequency = 250kHz, $\triangle Vo = 0.3$

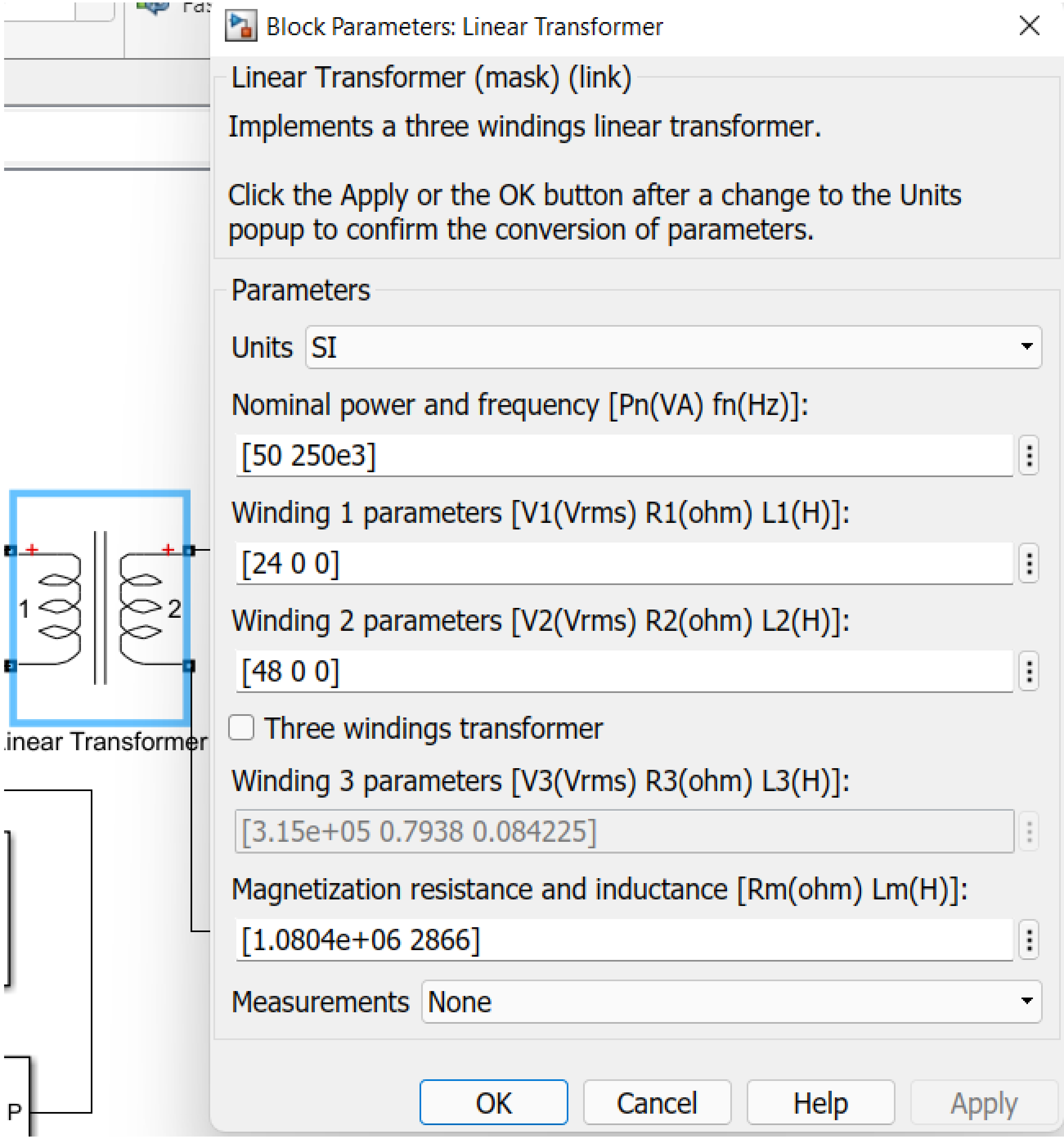
Duty Cycle

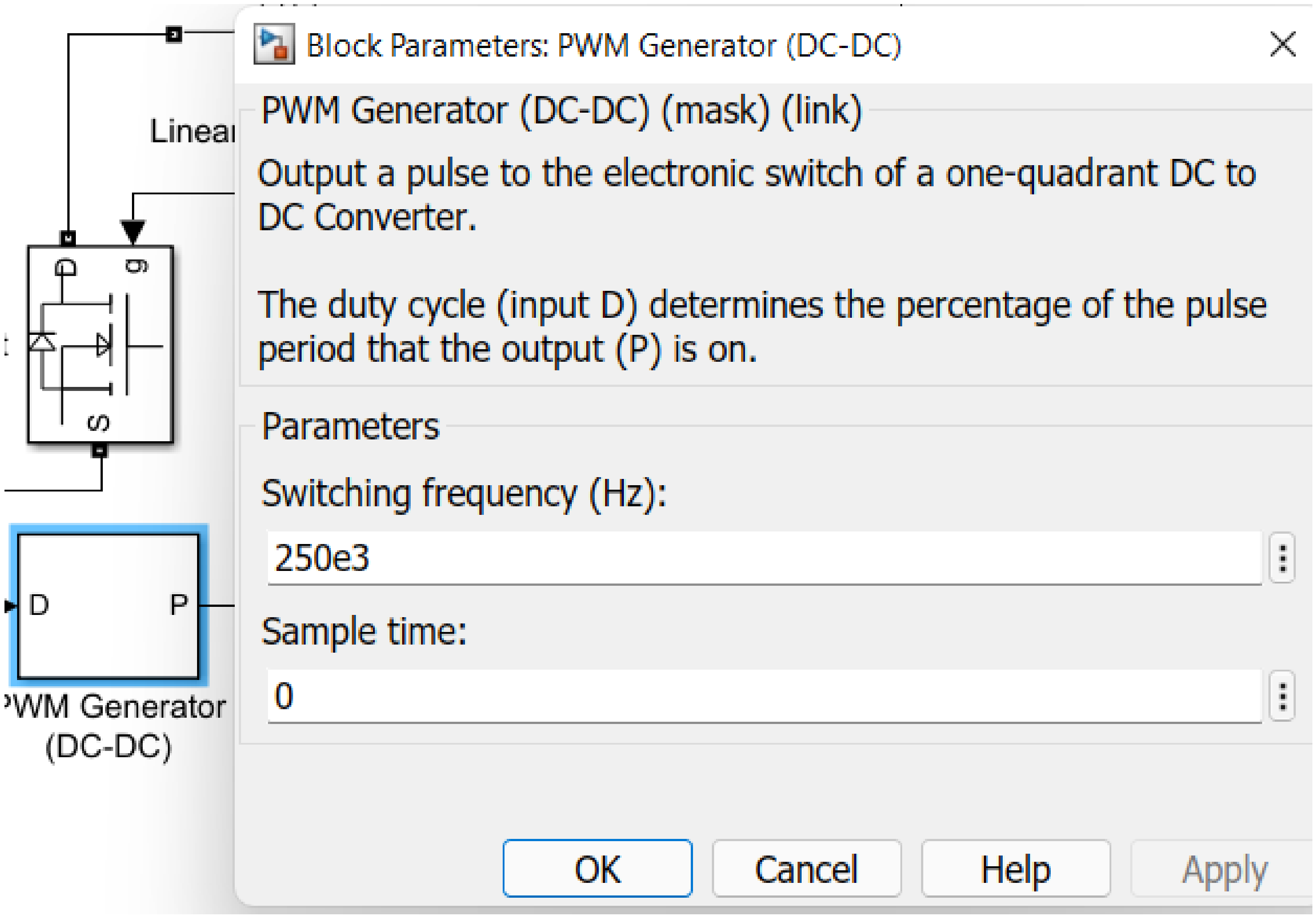
$$\frac{N_2}{N_1} = \frac{V_o}{V_s} * \frac{1-D}{D}$$

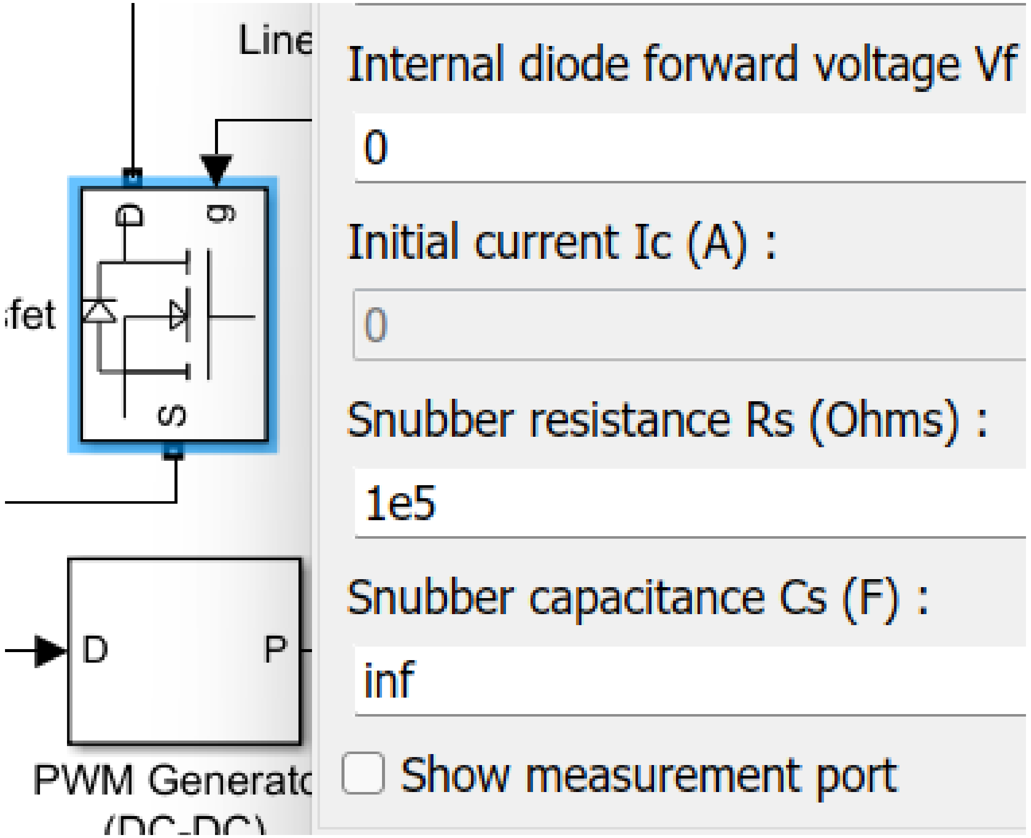
The turns ratio is 4 so the duty will be 0.3

$$C = \frac{D * Vo}{R * f * \triangle Vo}$$


Continuous
powergui
45.66
Display
Scope
m
a
k
Diode
1
2
Linear Transformer
Series RLC Branch1
Series RLC Branch
v
Voltage Measurement
RMS
RMS
45.66
Display1
DC Voltage Source
Mosfet
D
g
S
0.3
Constant
D
P
PWM Generator
(DC-DC)


Block Parameters: Linear Transformer
Linear Transformer (mask) (link)
Implements a three windings linear transformer.
Click the Apply or the OK button after a change to the Units popup to confirm the conversion of parameters.
Parameters
Units SI
Nominal power and frequency [Pn(VA) fn(Hz)]:
[50 250e3]
Winding 1 parameters [V1(Vrms) R1(ohm) L1(H)]:
[24 0 0]
Winding 2 parameters [V2(Vrms) R2(ohm) L2(H)]:
[48 0 0]
Three windings transformer
Winding 3 parameters [V3(Vrms) R3(ohm) L3(H)]:
[3.15e+05 0.7938 0.084225]
Magnetization resistance and inductance [Rm(ohm) Lm(H)]:
[1.0804e+06 2866]
Measurements None
OK
Cancel
Help
Apply
1
2
inear Transformer
P


Block Parameters: PWM Generator (DC-DC)
PWM Generator (DC-DC) (mask) (link)
Output a pulse to the electronic switch of a one-quadrant DC to DC Converter.
The duty cycle (input D) determines the percentage of the pulse period that the output (P) is on.
Parameters
Switching frequency (Hz):
250e3
Sample time:
0
OK
Cancel
Help
Apply
Linear
g
S
D
P
PWM Generator (DC-DC)



Line
fet
D
g
S
D
P
PWM Generato
(DC-DC)
Internal diode forward voltage Vf
0
Initial current Ic (A) :
0
Snubber resistance Rs (Ohms) :
1e5
Snubber capacitance Cs (F) :
inf
Show measurement port